

MONEYBALL

NEW SIZING RULES AND ALGORITHMS FOR UNMANNED LUNAR LANDERS

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I. Context – Why Size Unmanned Lunar Landers

II. Methods – Statistical Models and Sizing Algorithms

III. Results – Sizing Case Studies

IV. Conclusion

Context – The High Cost of Space Systems

A key variable: The Life Cycle Cost: $C_{LC} = C_{ddte} + C_p + C_o + C_d$



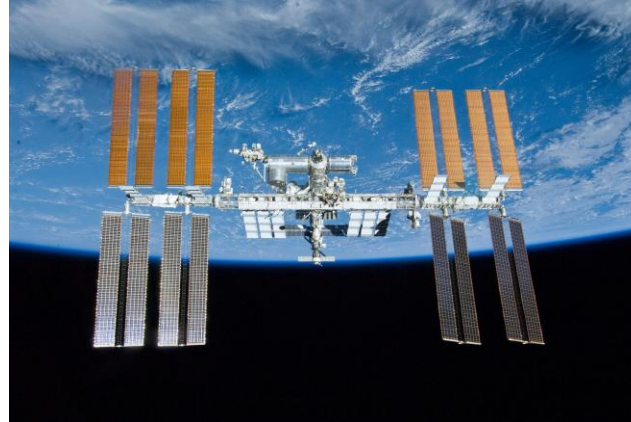
Harrison Schmitt in 1972

$$C_{LC}$$

257 Billion USD (2020) [1]

Samples 671 Million USD/kg

Earth Diamonds: 20.8M\$/kg[2]



International Space Station

$$C_{aq} = C_{ddte} + C_p$$

150 Billion USD (2010) [3]



Galileo Constellation

$$C_{aq} = C_{ddte} + C_p$$

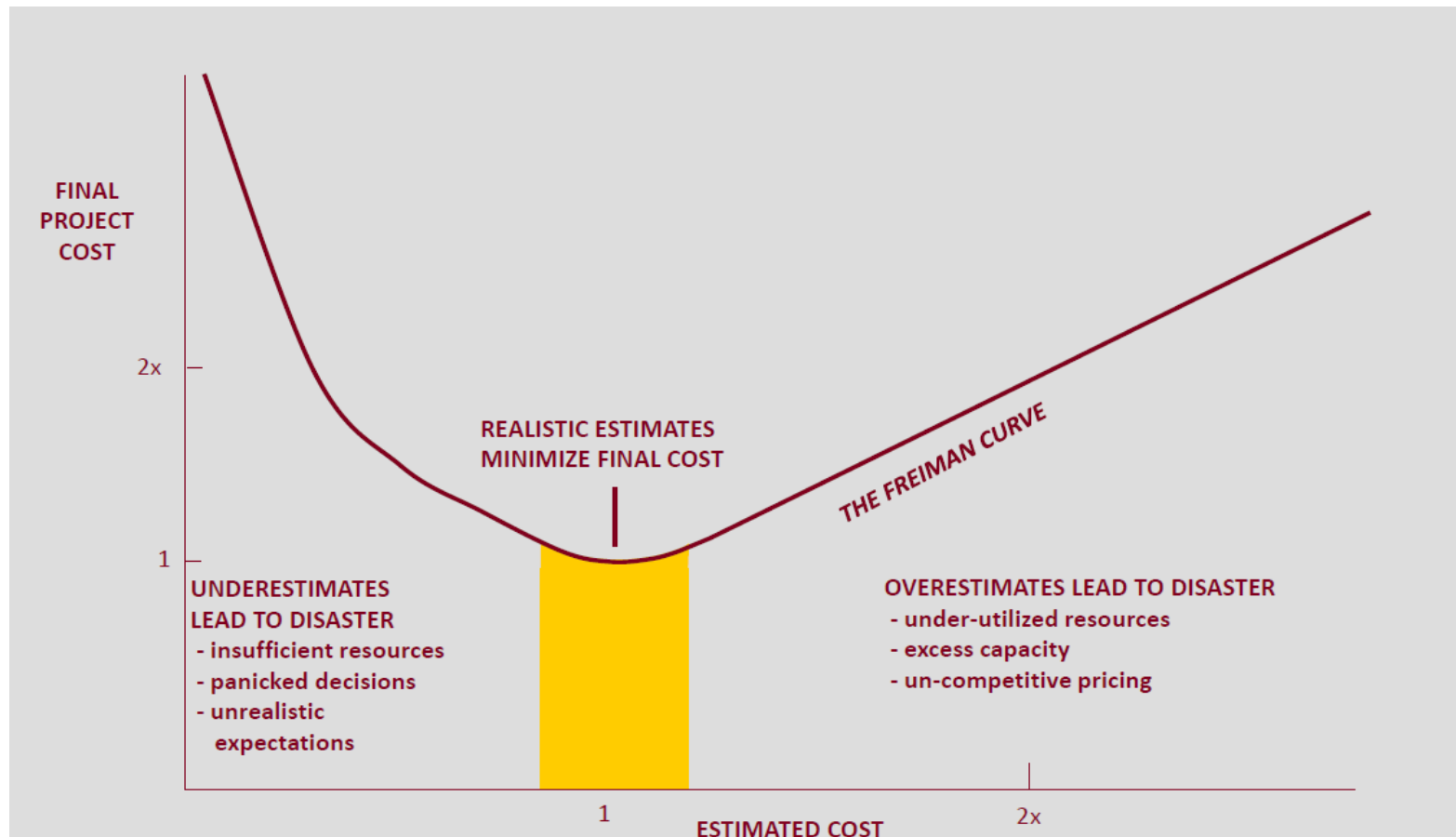
Block 1 : €953M (2010)[4]

Equivalent mass of gold: €462M [5]

To enable more like this, it needs to be cheaper

Context – The High Cost of Space Systems

A key to control costs: **Estimate**



Eilingsfeld F. Cost Engineering Lectures 2019 [3]

Order of Magnitude Estimation

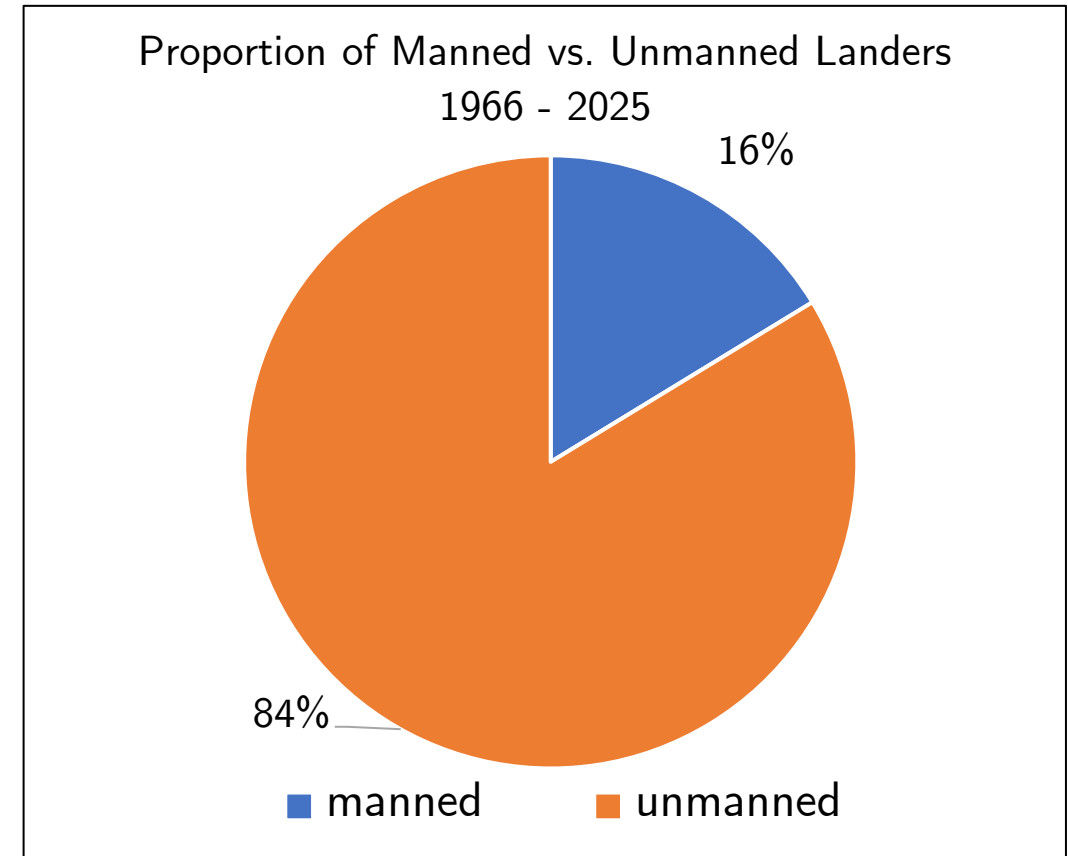
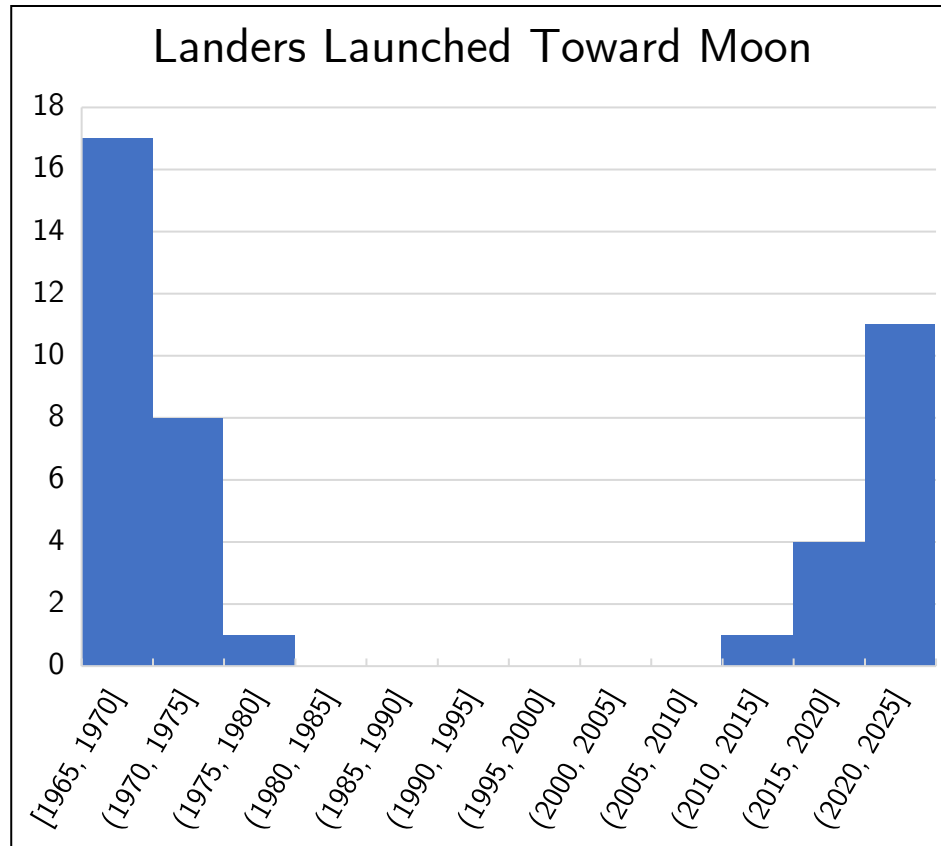
Component list

Parametric Cost Estimation Models (CEMs)

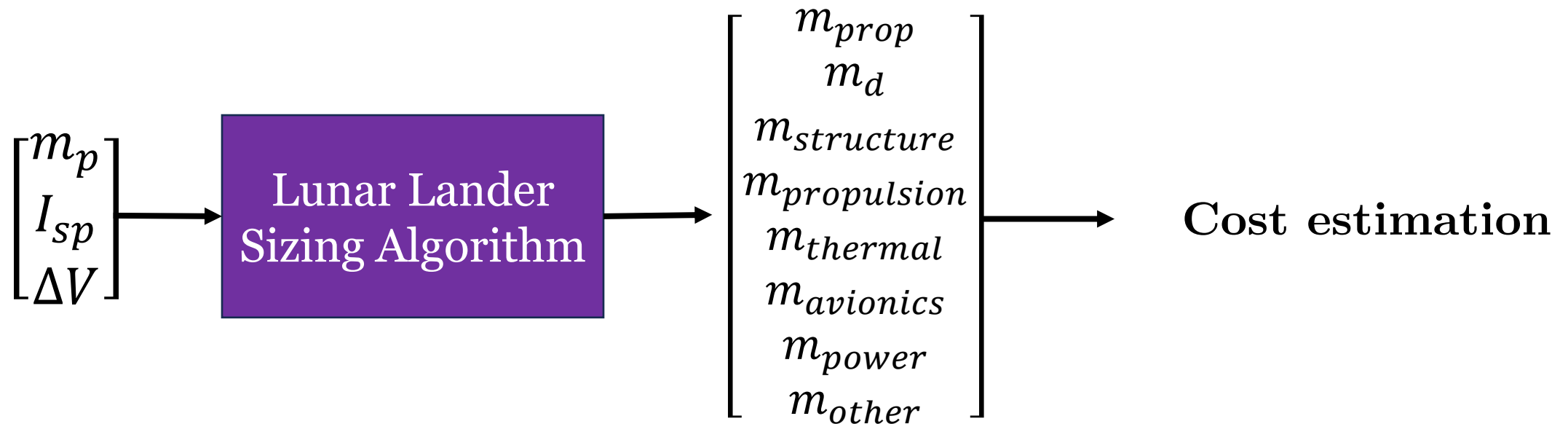
- QuickCost
- NAFCOM
- Advanced Mission Cost Model
- Small Spacecraft Cost Model
- TruePlanning

**All these CEMs require the
mass properties of the
(sub)system as an input**
Even better at subsystem level

Context – Why Study Unmanned Lunar Landers



Study Objective:



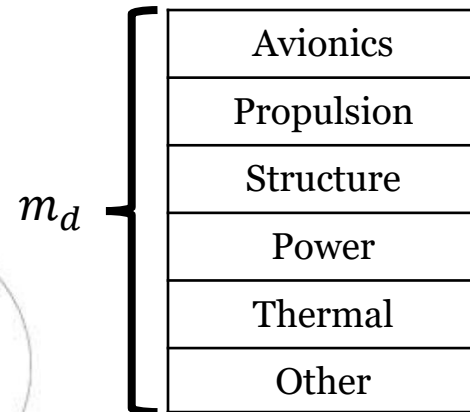
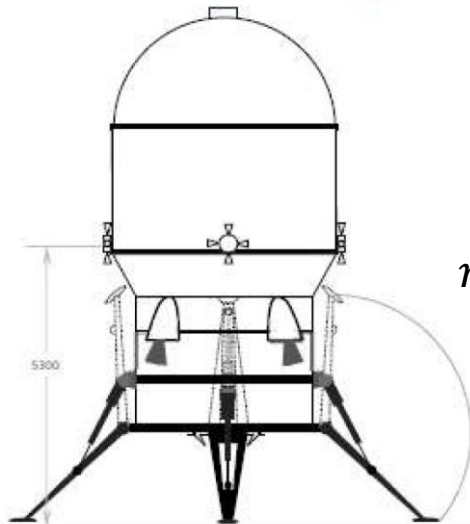
Study case: lunar missions

Application case: Lunar Surface Missions

How to approach this mass estimation problem: Reference Models

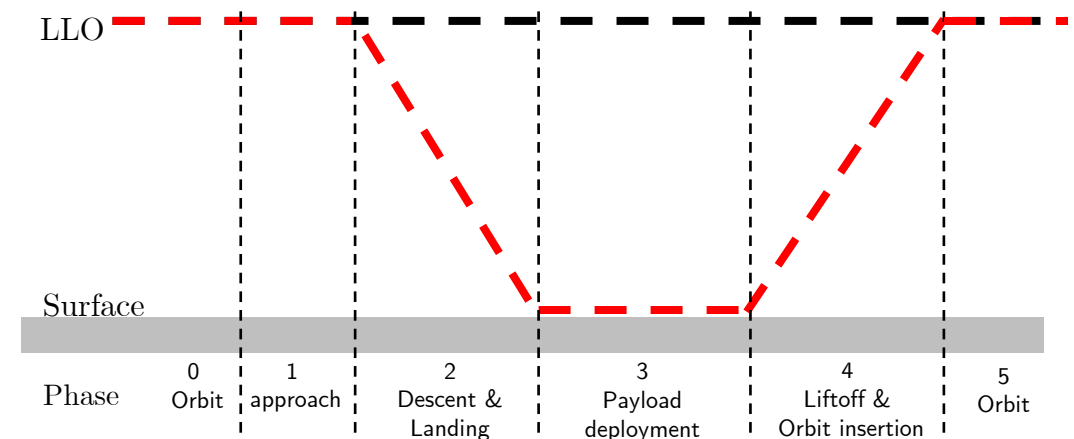
The Generic Lander Model

- Single stage
- Unmanned
- $m_t = m_p + m_d + m_{prop}$
- LOX-LH2, LOX-LCH4, N2O4-AZ

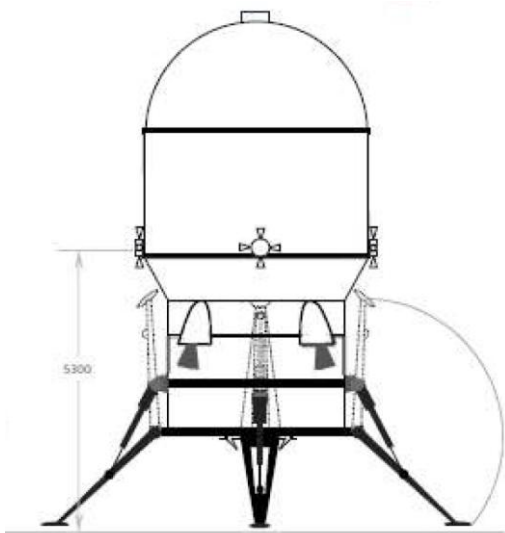


The Generic Mission Model

- Begin at LLO
- Land on the surface with all payload
- Return to LLO with some payload
- Sets the delta-V



First: for system level mass properties



$$\Delta V = I_{sp} g_0 \ln\left(\frac{m_i}{m_f}\right)$$

$$m_i = m_t$$

$$m_f = m_p + m_d$$

known

$$m_{prop} = (m_d + m_p) \cdot \left(e^{\frac{\Delta V}{I_{sp} g_0}} - 1\right)$$

$$m_t = m_p + m_d + m_{prop}$$

Unknown

Solution: find additional relations using statistics

System masses DB (#35)
(Almost) all flight models

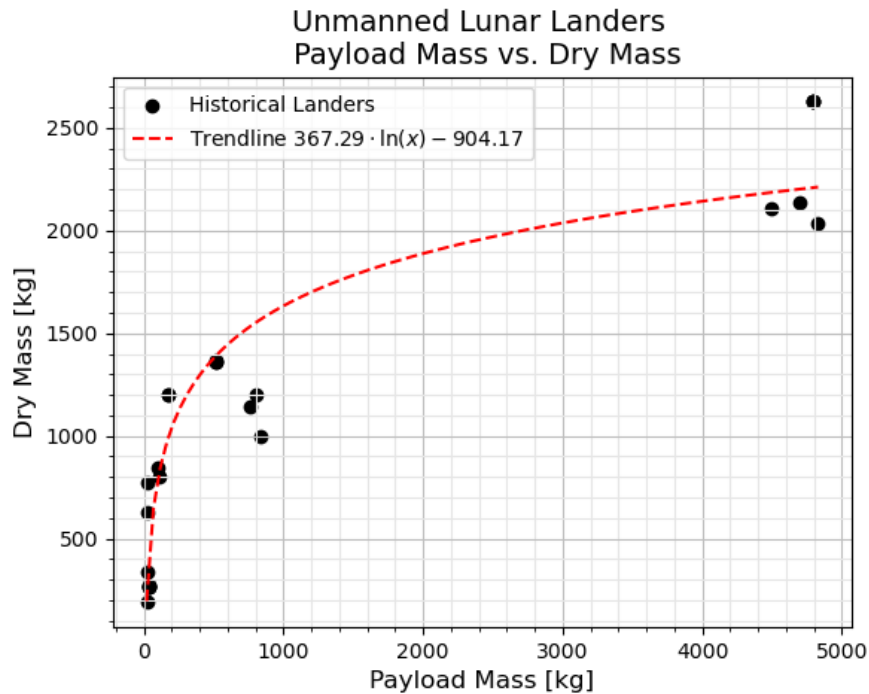
Year	Organisation	Spacecraft name	Mission	mt	mp	md	mprop	source
1966	USSR	Luna 9	Ye-6M	1538	99.8	847	591.2	14,24,44
1966	NASA	Surveyor 1	Surveyor 1	995.2	33	262	700.2	33,38
1966	USSR	Luna 13	Ye-6M	1620	112	800	708	13,14,23
1967	NASA	Surveyor 5	Surveyor 5	1006	33	271	702	35,38
1967	NASA	Surveyor 6	Surveyor 6	1026	33	267	726	36,38
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
2024	Intuitive Machines	Odysseus	IM-1	1908	130	494	1276	5,15,29
2024	CNSA	Chang'e 6	Chang'e 6	3800	800	1200	1800	28,40,48
2025	Firefly	Blueghost	Blueghost M1	1517	150	331	1036	2,6,7
2025	Intuitive Machines	Athena	Athena	2120	130	545	1445	5,11,16
2025	iSpace	Hakuto-R M2	Hakuto-R M2	1000	30	340	630	25

Subsystem masses DB (#12)
Flown landers and concepts
(payload for manned landers is ascent stage)

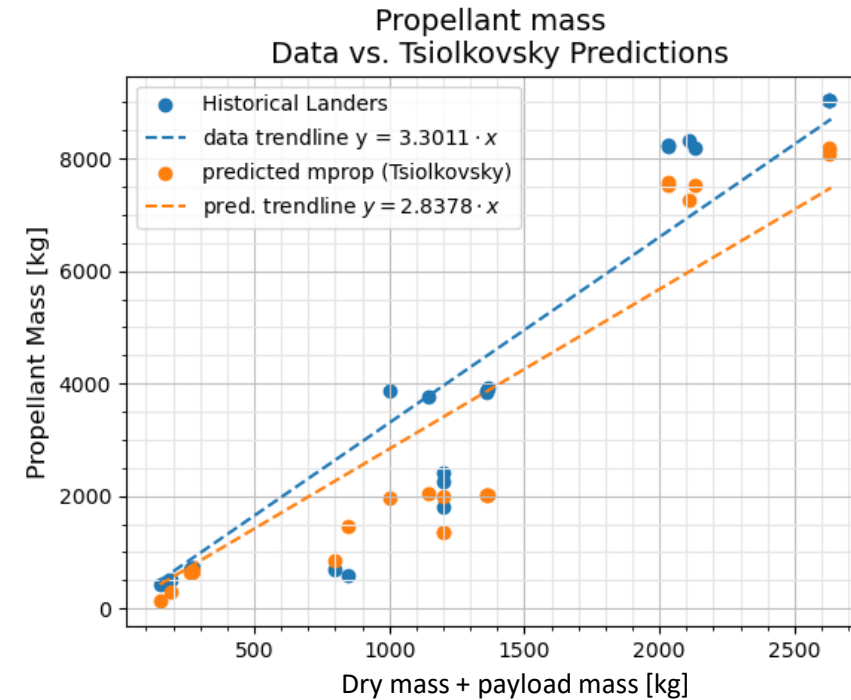
Lander	md	mp	mprop	dV	Isp	Structure	Propulsion	Power	Avionics	Thermal	Other
Apollo J-series	2373	5295	8780	2265	311	460	495	366	29	404	273
9508-HLR-1	1673.7	473.3	2891.6	1822	333	532.9	252.5	125.7	120.1	113.5	56
8801-EE-1 LOX-LH2	9823	25000	25252	2100	450	1681	4258	478	934	2017	455
8801-EE-1 N2O4/MMH	7899	25000	36399	2100	333	1955	2943	478	1062	1006	455
9205-FLO-1-CLS	9320	35894	44151	3100	450	1770	4969	385	307	420	1469
0503-CE&R-8	5858	16461	22502	2260	379	2822	1749	204	818	265	0
0609-LLPS-GRC	6503	229	4448	ND	450	1186.34	918.21	663	258.7	459.04	0
0610-LAT-1	10747	22074	50596	ND	450	453	2969	288	519	56	527
0507-ESAS-D	7542	28475	25486	1900	362	2015	2425	553	594	1072	883
0507-ESAS-F	7412	26375	23833	1900	351	1985	2347	553	594	1066	867
0507-ESAS-H	5970	22028	19152	1900	363	1877	1125	553	594	1049	772
0507-ESAS-J	5912	20956	18312	1900	363	1860	1095	553	594	1046	764

Results – Sizing Rules

For the system masses: m_d , and m_{prop}



$$m_d = 367.29 \cdot \ln(m_p) - 904.17$$
$$R^2 = 0.92$$



$$m_{prop} = 1.16 \cdot (m_d + m_p) \cdot \left(e^{\frac{\Delta V}{I_{sp} g_0}} - 1 \right)$$

For the subsystem masses: multiple linear regression, and bottom up sizing

$$m_{ss} = \begin{bmatrix} m_d \\ m_p \\ m_{prop} \\ I_{sp} \\ \Delta V \\ 1 \end{bmatrix} [a, b, c, d, e, intercept]$$

Component level

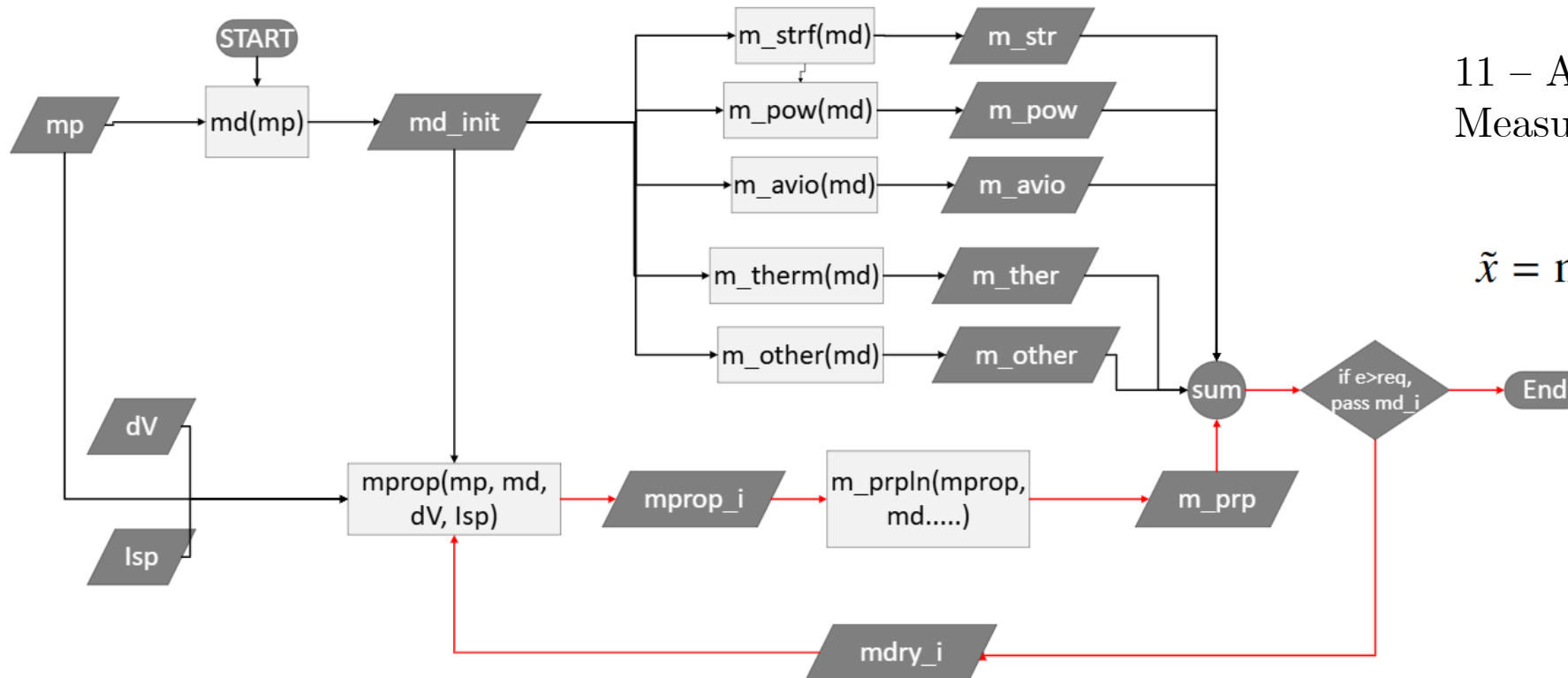
$$m_{prpn} = m_{engines} + m_{tanks} + m_{plumbing}$$

Subsystem	a	b	c	d	e	intercept	R ²
Structure	-0.2100	0.0172	0.0746	-1.5635	7.9994	1,356	0.5431
Propulsion	0.6449	-0.0645	0.0100	1.4171	1.1680	-4,288	0.9547
Power	0.0818	0.0217	-0.0241	0.1914	-3.5158	897	0.8619
Avionics	0.1108	-0.0415	0.0406	-0.8425	1.0197	1,207	0.8728
Thermal	0.5412	-0.0130	-0.0714	0.2366	-6.7381	1,269	0.8976
Other	-0.1543	0.0910	-0.0276	0.4351	0.0745	-617	0.8947

engine	tanks	plumbing
Thrust	tank volume number of tanks tank material internal pressure	15% of mass of subsystem

Problems with outliers: Single linear regression used also

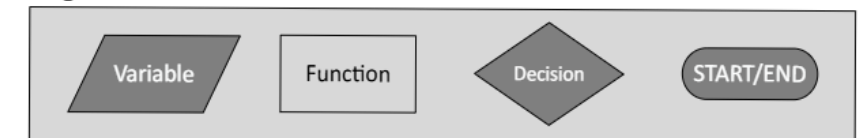
Sizing Algorithm



11 – Algorithms tested
Measured by median absolute error

$$\tilde{x} = \text{median} \left(\left\{ \frac{\|\hat{y}_i - y_i\|}{y_i} \right\}_{i=1}^n \right)$$

Legend



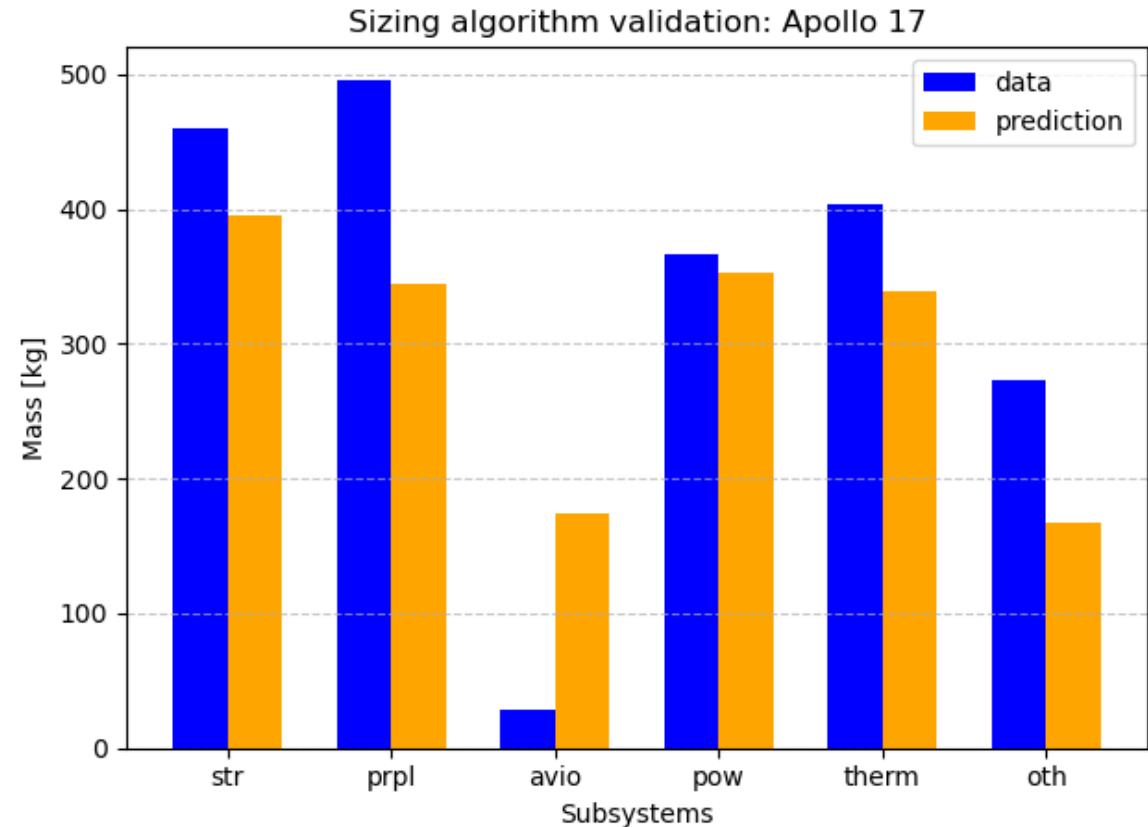
Results – Validation

Resizing of the J-series Apollo LM.
Algo No. 7 N2O4/AZ

$$\begin{bmatrix} m_p \\ I_{sp} \\ \Delta V \end{bmatrix} = \begin{bmatrix} 5295 \\ 311 \\ 2265 \end{bmatrix}$$

all LM values from *Apollo by the Numbers*
(Orloff 1996)

Good estimation of system level masses
Some large errors in ss sizing



Resizing all landers in the DB. The best performers: all single linear regressors

Mass property	N2O4 Algo 7 (%)	LOX/LH2 Algo 10 (%)	LOX/LCH4 Algo 11 (%)
m_p	0.00	0.00	0.00
m_t	3.63	13.29	5.51
m_d	28.67	31.02	34.855
m_{prop}	13.19	17.43	16.54
m_{str}	13.17	13.17	13.17
m_{prpn}	68.36	79.81	80.425
m_{avio}	64.04	53.04	66.59
m_{pow}	34.37	26.85	34.485
m_{ther}	17.56	21.49	24.79
m_{oth}	30.75	30.75	30.75

Contributions:

- Most up to date published lunar lander database used for sizing rule derivation.
- Algorithms to size lunar landers to the subsystem level using only m_p , I_{sp} and ΔV .

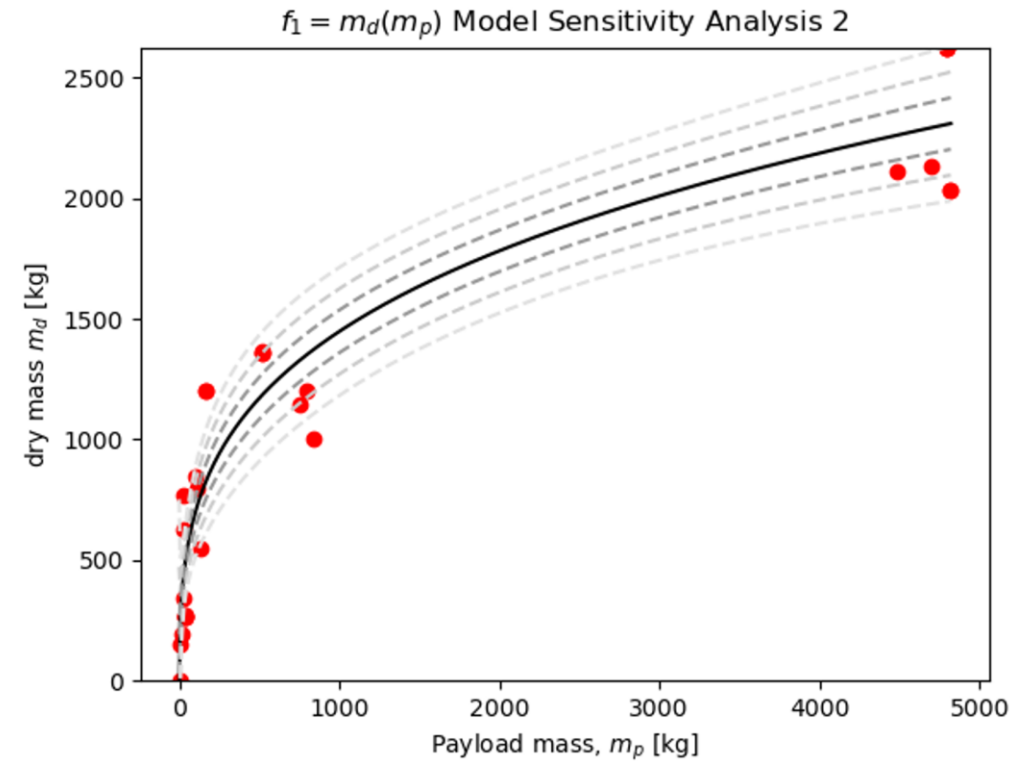
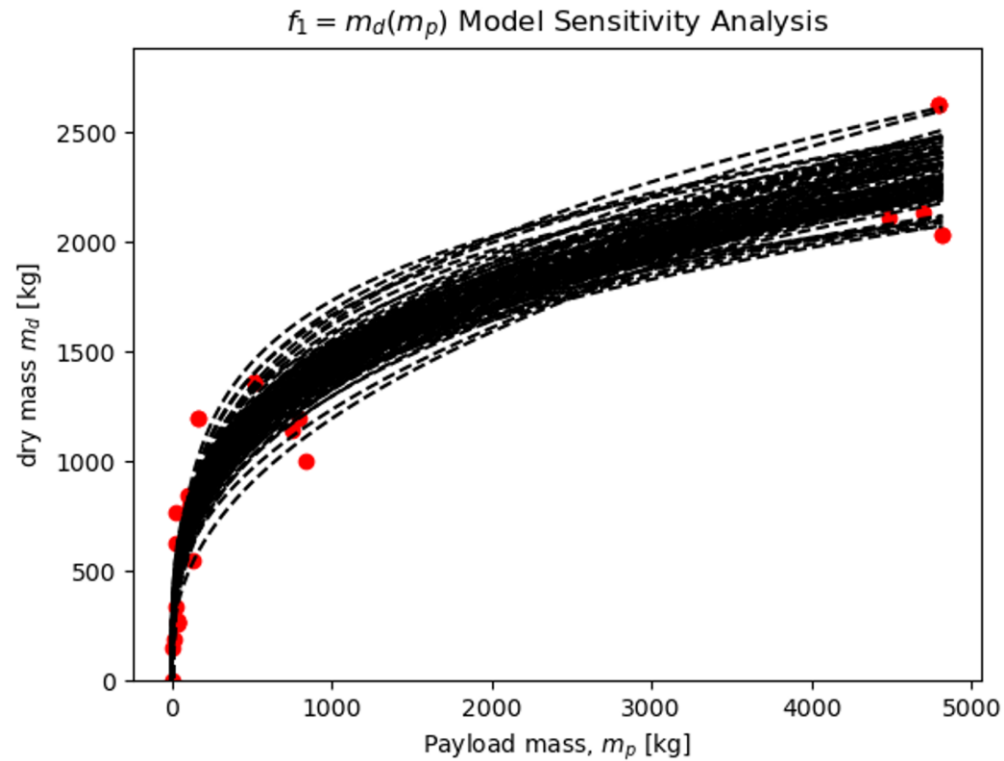
Future Work

- Must improve subsystem mass estimates
- Uncertainty propagation through the sizing algorithm and progressive estimation starting with the highest R^2 could improve subsystem mass estimates.

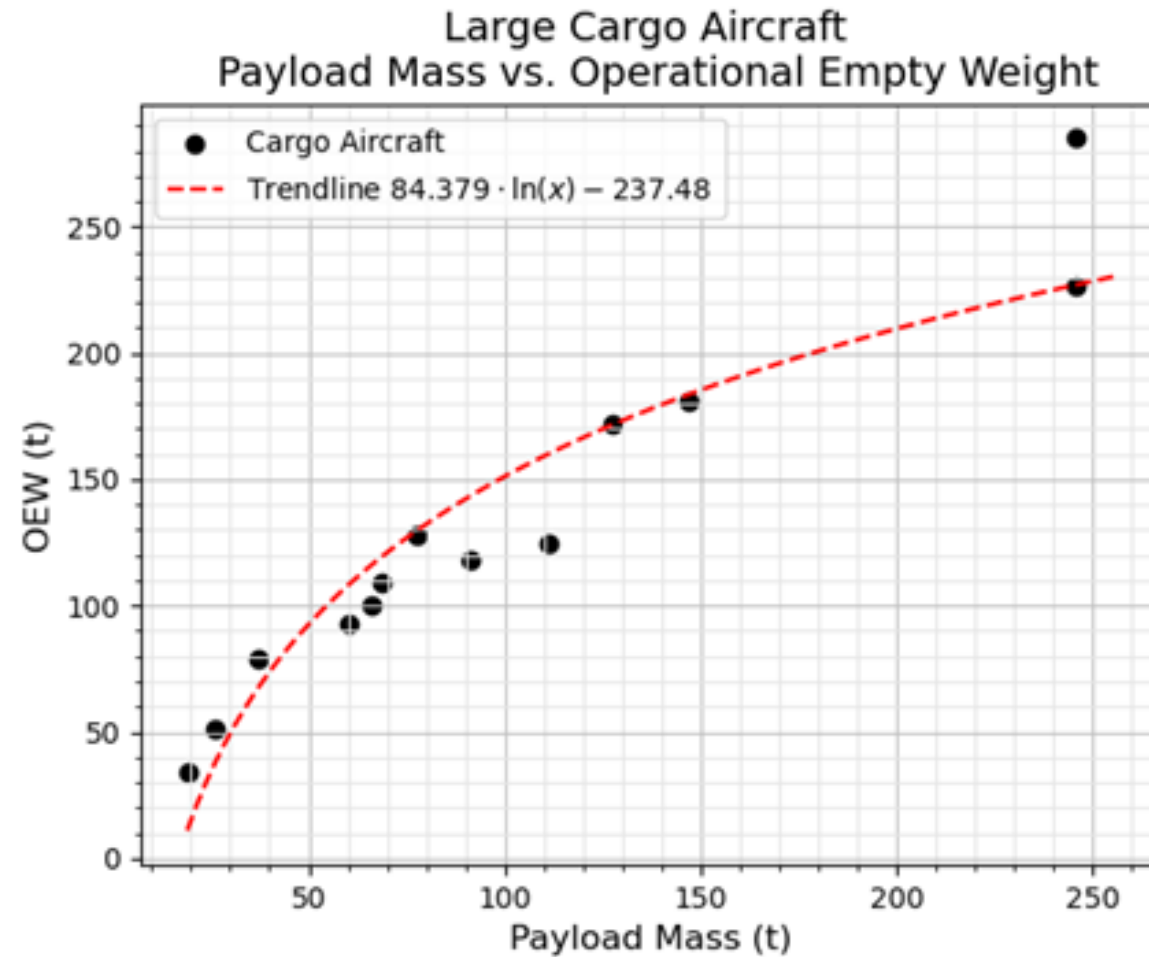
1. Amos, J., 2010. EU awards Galileo satellite-navigation contracts. BBC News, 7 January. Available at: <http://news.bbc.co.uk/2/hi/science/nature/8442090.stm>
2. StoneAlgo (2024) Diamond Prices – Real-Time Market Trends. Available at: <https://www.stonealgo.com/diamond-prices/?srsltid=AfmBOooDezL1x8X3j9bA7wr0FFT2uVo1hgZrP4pOa6aAj3RQ3cClizXn> (Accessed: 28 May 2025).
3. Lafleur, C., 2010. Costs of US piloted programs. The Space Review, 8 March. Available at: <http://www.thespacereview.com/article/1579/1>
4. Amos, J., 2010. EU awards Galileo satellite-navigation contracts. BBC News, 7 January. Available at: <http://news.bbc.co.uk/2/hi/science/nature/8442090.stm>
5. GoldPrice.org, 2025. Gold Price Charts & Historical Data [online] Available at: <https://www.goldprice.org/>
6. F. Eilingsfeld, 2019 Introduction to Cost Engineering lecture 1.

Thank you !

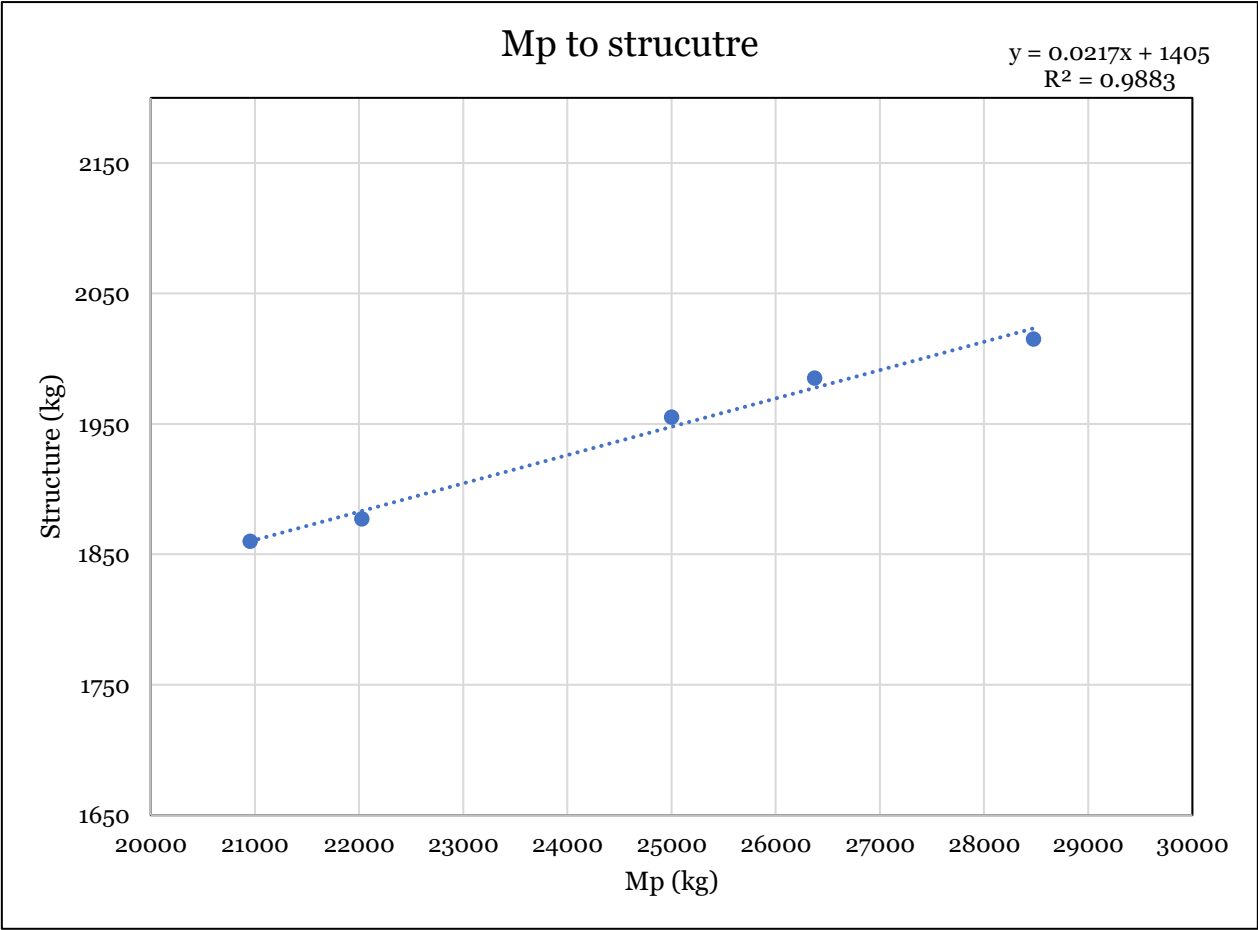
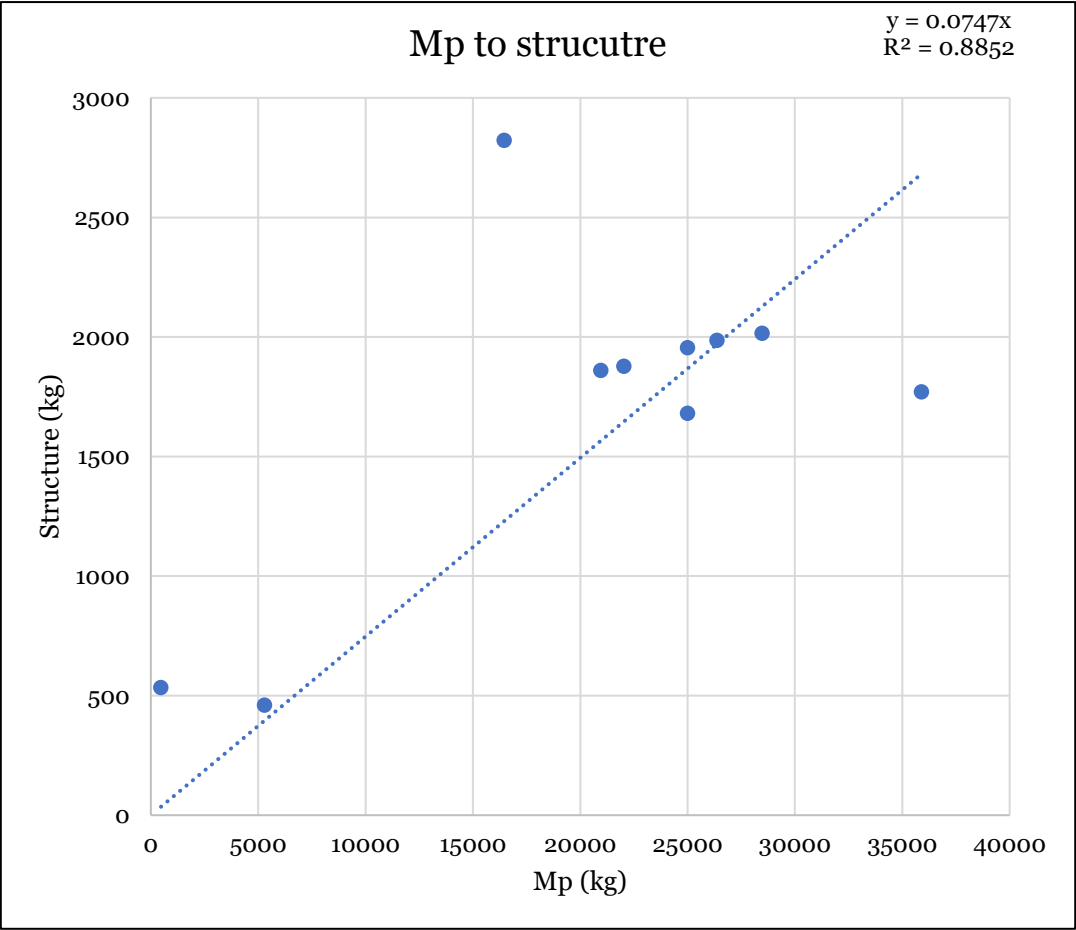
Extra slides – Uncertainty



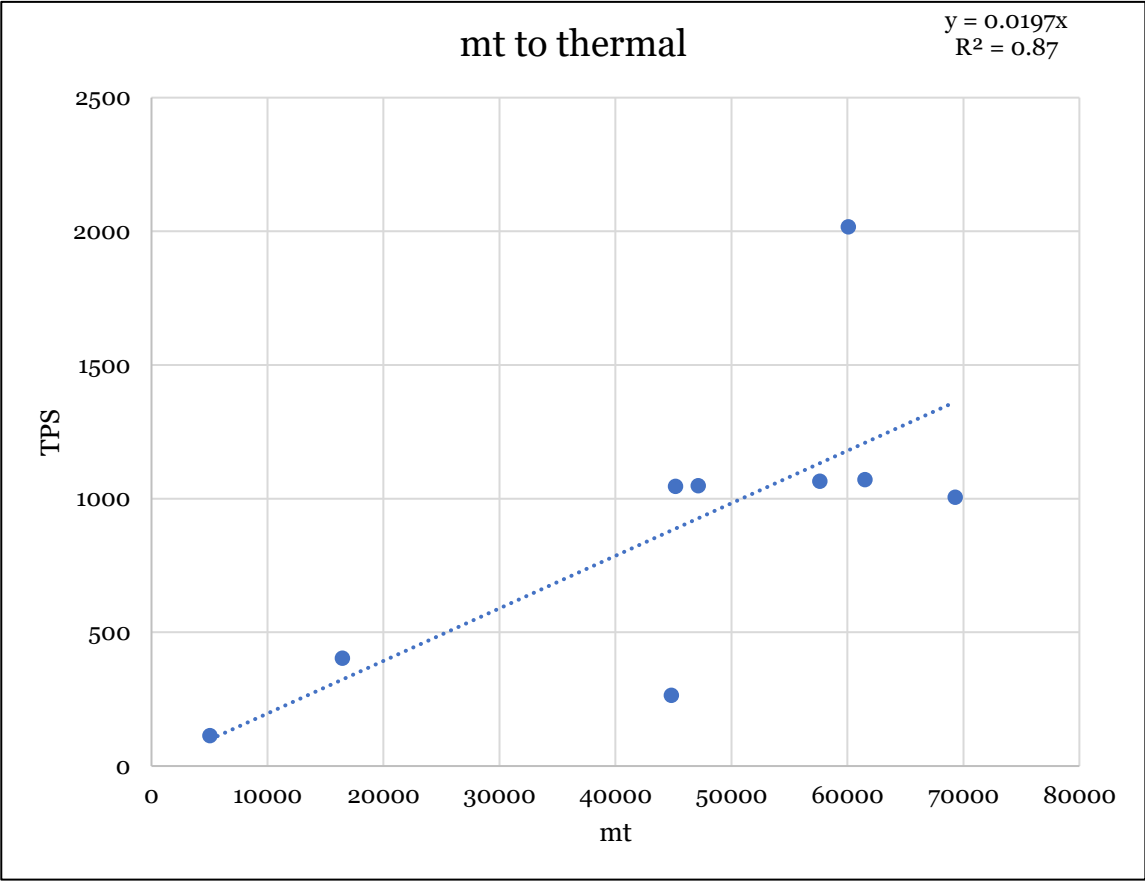
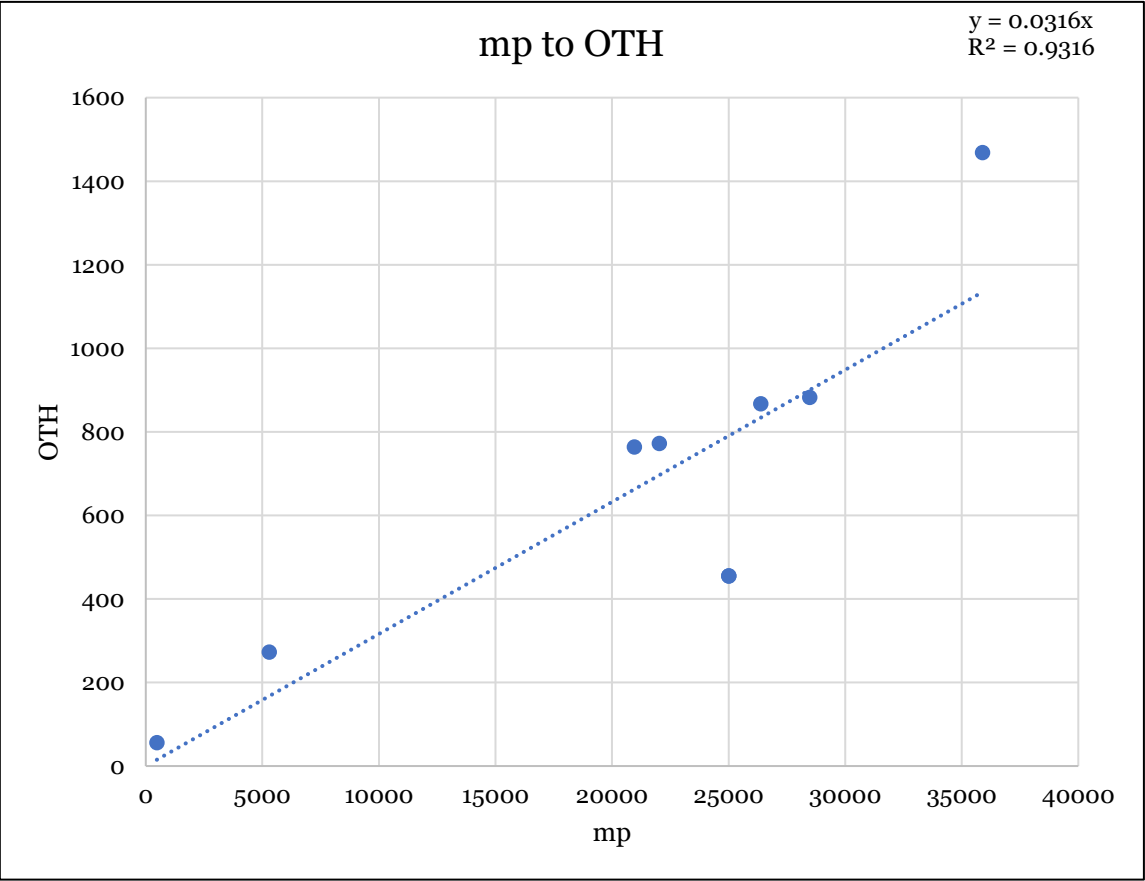
Extra slides Cargo Aircraft



Extra Single Linear Regression



Extra Single Linear Regression



Extra Single Linear Regression

